rocks condensed out of lava between 180 million and 1.3 billion years ago. Then they were driven off the planet by collisions from space. From how long they've been exposed to cosmic rays on their interplanetary journeys between Mars and Earth, we can tell how old they are—how long ago they were ejected from Mars. In this sense, they are between 10 million and 700,000 years old. They sample the most recent 0.1 percent of Martian history.

Some of the minerals they contain show clear evidence of having once been in water, warm liquid water. These hydro-thermal minerals reveal that somehow, probably all over Mars, there was recent liquid water. Perhaps it came about when the interior heat melted underground ice. But however it happened, it's natural to wonder if life is not entirely extinct, if somehow it's managed to hang on into our time in transient underground lakes, or even in thin films of water wetting subsurface grains.

The geochemists Everett Gibson and Hal Karlsson of NASA's Johnson Space Flight Center have extracted a single drop of water from one of the SNC meteorites. The isotopic ratios of the oxygen and hydrogen atoms that it contains are literally unearthly. I look on this water from another world as an encouragement for future explorers and settlers.

Imagine what we might find if a large number of samples, including never melted soil and rocks, were returned to Earth from Martian locales selected for their scientific interest. We are very close to being able to accomplish this with small roving robot vehicles.

The transportation of subsurface material from world to world raises a tantalizing question: Four billion years ago there were two neighboring planets, both warm, both wet. Impacts from space, in the final stages of the accretion of these planets, were occurring at a much higher rate than today. Samples from each world were being flung out into space. We are sure there was life on at least one of them in this period. We know that a fraction of the ejected debris stays cool throughout the processes of impact, ejection, and interception by another world. So could some of the early organisms on Earth have been safely transplanted to Mars four billion years